Problem 11: A Matcha a Day
15 Points

Problem ID: `matcha`
Rank: `4`

Introduction

Big Ben is a big matcha fan, and he is committed to getting a cup of matcha from a new matcha store every day of his life. Unfortunately, his life is finite (or so it seems...), and he needs your help to devise a plan of ordering matcha to maximize his matcha enjoyment over his life!



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Problem Statement

Big Ben's life spans $\mathbf{N}$ days, numbered $1, 2, \ldots, \mathbf{N}$. On day i, he visits a matcha store and must choose an integer f_i denoting his chosen matcha flavor using the range $\mathbf{L}_i \leq f_i \leq \mathbf{R}_i$ offered by that store. Big Ben's enjoyment is defined as the bitwise XOR of his chosen flavors, $f_1 \oplus f_2 \oplus \cdots \oplus f_{\mathbf{N}}$.

Given $\mathbf{L}_1, \mathbf{L}_2, \ldots, \mathbf{L}_{\mathbf{N}}$ and $\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_{\mathbf{N}}$, find the maximum possible enjoyment Big Ben can achieve over all ways of choosing $f_1, f_2, \ldots, f_{\mathbf{N}}$.

Input Format

The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases that follow. For each test case:

- The first line contains a single integer $\mathbf{N}$ denoting the number of days in Big Ben's life.
- The second line contains $\mathbf{N}$ space-separated integers $\mathbf{L}_1\ \mathbf{L}_2\ \ldots\ \mathbf{L}_{\mathbf{N}}$.
- The third line contains $\mathbf{N}$ space-separated integers $\mathbf{R}_1\ \mathbf{R}_2\ \ldots\ \mathbf{R}_{\mathbf{N}}$.

Output Format

For each test case, output a single line containing a single integer, the maximum possible enjoyment.

Constraints

$1 \leq \mathbf{T} \leq 10^5$
$1 \leq \mathbf{N} \leq 2 \cdot 10^5$
$0 \leq \mathbf{L}_i \leq \mathbf{R}_i < 2^{30}$
The sum of $\mathbf{N}$ across all test cases in an input file does not exceed $2 \cdot 10^5$.

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Sample Test Cases

Sample Input Download

```
4
1
6
7
3
0 0 0
1073741823 1073741823 1073741823
4
0 1 3 5
1 2 4 8
3
98 98 101
105 103 110
```

Sample Output Download

```
7
1073741823
15
111
```

Sample Explanations

For the first test case: There is one matcha store, characterized by $\mathbf{L}_1 = 6$ and $\mathbf{R}_1 = 7$. Big Ben's can choose $f_1 = 7$, leading to an enjoyment of 7. This can be shown to be the maximum happiness possible.

For the second test case: Big Ben can choose $f_1 = 0$, $f_2 = 0$, and $f_3 = 1073741823$, leading to an enjoyment of $0 \oplus 0 \oplus 1073741823$. This can be shown to be the maximum happiness possible.

For the third test case: Big Ben can choose $f_1 = 1$, $f_2 = 2$, $f_3 = 4$, and $f_4 = 8$, leading to an enjoyment of $1 \oplus 2 \oplus 4 \oplus 8 = 15$. This can be shown to be the maximum happiness possible.

For the fourth test case: Big Ben can choose $f_1 = 98$, $f_2 = 99$, and $f_3 = 110$, leading to an enjoyment of $98 \oplus 99 \oplus 110 = 111$. This can be shown to be the maximum happiness possible.

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